```
============================================================
A.1 - Single DES run for lambda = 100 req/sec
============================================================
Simulated 10000 fully served requests up to t = 100837.6373 ms = 100.8376 sec
Total arrivals observed:        10002
R1, type-1 phase response:      10.579663 ms
R2, type-2 phase response:      6.398254 ms
R end-to-end:                   16.977038 ms
Type-1 jobs served per second:  99.179238
Type-2 jobs served per second:  99.169321
Mean #type-1 in system, tw:     1.049398
Mean #type-2 in system, tw:     0.634528
Mean total #jobs in system, tw: 1.683926

============================================================
A.2 - Stationarity across lambda values
============================================================
E[S1] = exp(1.5) = 4.481689 ms
E[S2] = 0.750000 ms
E[S1] + E[S2] = 5.231689 ms
Analytical stability condition: rho = lambda*(E[S1]+E[S2])/1000 < 1
Stability bound: lambda < 191.142858 req/sec
Running lambda = 50 req/sec, rho = 0.261584 -> stable ... done, t_end = 203.2197
sec, arrivals =
 10000, served = 10000
Running lambda = 100 req/sec, rho = 0.523169 -> stable ... done, t_end =
100.0762 sec, arrivals
= 10000, served = 10000
Running lambda = 150 req/sec, rho = 0.784753 -> stable ... done, t_end = 67.1597
sec, arrivals =
 10002, served = 10000
Running lambda = 200 req/sec, rho = 1.046338 -> unstable ... done, t_end =
52.5433 sec, arrivals
 = 10377, served = 10000
Running lambda = 250 req/sec, rho = 1.307922 -> unstable ... done, t_end =
65.0984 sec, arrivals
 = 16152, served = 10000

Analytical classification for A.2:
  lambda =  50 req/sec: rho = 0.261584 < 1 -> stationary
  lambda = 100 req/sec: rho = 0.523169 < 1 -> stationary
  lambda = 150 req/sec: rho = 0.784753 < 1 -> stationary
  lambda = 200 req/sec: rho = 1.046338 >= 1 -> non-stationary
  lambda = 250 req/sec: rho = 1.307922 >= 1 -> non-stationary

============================================================
A.3 - Confidence intervals from independent replications
============================================================

lambda = 100 req/sec: running 30 independent
replications .............................. done
  Full trace, type 1: mean CI [1.014991, 1.056171], median CI [1.011125,
1.063644]
  Full trace, type 2: mean CI [0.603187, 0.632896], median CI [0.601051,
0.631790]
  After transient removal, type 1: mean CI [1.008903, 1.060815], median CI
[0.984703, 1.060240]
  After transient removal, type 2: mean CI [0.599920, 0.636938], median CI
[0.586471, 0.647216]

lambda = 180 req/sec: running 30 independent
replications .............................. done
  Full trace, type 1: mean CI [14.476681, 18.489433], median CI [13.566229,
18.068241]
```

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  Full trace, type 2: mean CI [13.679712, 17.688760], median CI [12.826022,
17.198553]
  After transient removal, type 1: mean CI [14.638395, 19.446088], median CI
[13.488622, 18.9447
85]
  After transient removal, type 2: mean CI [13.891899, 18.707440], median CI
[12.648005, 18.3375
82]

============================================================
A.4 - Little's Law verification for lambda = 100 req/sec
============================================================
Without transient removal:
  R = 16.977038 ms
  N_sim, time-weighted total jobs = 1.683926
  lambda * R = (0.100000 req/ms) * 16.977038 ms = 1.697704
  Percent error = 0.818167 %

With transient removal:
  Cutoff = 30251.291177 ms = 30.251291 sec, 30% of run time
  R = 16.830559 ms
  N_sim, time-weighted total jobs after cutoff = 1.651705
  lambda * R = (0.100000 req/ms) * 16.830559 ms = 1.683056
  Percent error = 1.898066 %

============================================================
FINAL SUMMARY - Lab 5, Part A
============================================================

A.1  lambda = 100 req/sec, 10000 fully served requests
  t_end = 100837.637255 ms = 100.837637 sec
  arrivals = 10002, fully served = 10000
  R1 = 10.579663 ms, R2 = 6.398254 ms, R = 16.977038 ms
  throughput: type-1 = 99.179238 jobs/sec, type-2 = 99.169321 jobs/sec
  time-weighted mean jobs: N1 = 1.049398, N2 = 0.634528, N = 1.683926

A.2  Stability bound
  lambda < 191.142858 req/sec
  lambda =  50: rho = 0.261584 -> stationary
  lambda = 100: rho = 0.523169 -> stationary
  lambda = 150: rho = 0.784753 -> stationary
  lambda = 200: rho = 1.046338 -> non-stationary
  lambda = 250: rho = 1.307922 -> non-stationary

A.3  95% confidence intervals, n = 30 replications
  Transient removal rule: remove first 30% of each run, guided by MQS(t) plots.

  lambda = 100 req/sec
    Full trace:
      type 1: mean CI [1.014991, 1.056171], median CI [1.011125, 1.063644]
      type 2: mean CI [0.603187, 0.632896], median CI [0.601051, 0.631790]
    After transient removal:
      type 1: mean CI [1.008903, 1.060815], median CI [0.984703, 1.060240]
      type 2: mean CI [0.599920, 0.636938], median CI [0.586471, 0.647216]

  lambda = 180 req/sec
    Full trace:
      type 1: mean CI [14.476681, 18.489433], median CI [13.566229, 18.068241]
      type 2: mean CI [13.679712, 17.688760], median CI [12.826022, 17.198553]
    After transient removal:
      type 1: mean CI [14.638395, 19.446088], median CI [13.488622, 18.944785]
      type 2: mean CI [13.891899, 18.707440], median CI [12.648005, 18.337582]

A.4  Little's Law, lambda = 100 req/sec
```

```
  Without transient removal:
    R = 16.977038 ms, N_sim = 1.683926, lambda*R = 1.697704, error = 0.818167 %
  With transient removal:
    R = 16.830559 ms, N_sim = 1.651705, lambda*R = 1.683056, error = 1.898066 %

Done.
```